

Asia Session Sweep System:

A Quantitative Framework for Institutional Liquidity Trading in Equity Index Futures

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Abstract

This paper presents the complete theoretical foundation, technical architecture, and empirical validation of the Asia Session Sweep System, an automated signal detection and risk management framework for day trading equity index futures. The system is built upon the institutional liquidity theory popularized by ICT (Inner Circle Trader) and operationalized through a nine-point confluence scoring model, a four-layer adaptive market context filter, and a persistent machine-learning-adjacent memory module that continuously calibrates thresholds from observed trade outcomes. A 60-day backtest over March through May 2026 MNQ data produced an 88.9 percent win rate across nine qualifying trades following three rounds of filter iteration. The system's selectivity is a deliberate design choice: the majority of potential setups are rejected by the filter stack, and only setups where every condition aligns are executed. This paper documents every layer of the system in sufficient detail to reproduce, audit, and extend it.

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1. Introduction and Motivation

The retail futures trading market is characterized by a fundamental structural asymmetry: large institutional participants possess informational, capital, and technological advantages that allow them to systematically profit at the expense of undercapitalized retail traders. This paper describes a trading system designed to detect and follow institutional order flow rather than compete against it. The underlying hypothesis is that institutional participants leave identifiable footprints in price structure that, when interpreted correctly through the lens of liquidity theory, provide statistically reliable entry signals with defined risk parameters.

The system targets MNQ (Micro E-mini Nasdaq-100) futures traded on the Chicago Mercantile Exchange. MNQ provides a dollar value of two dollars per index point per contract, making it accessible for evaluation accounts in the range of twenty-five thousand dollars. The system is designed to operate within the strict risk parameters of proprietary trading firm evaluations, specifically the Tradeify platform, which imposes a one-thousand-dollar trailing maximum drawdown and a fifteen-hundred-dollar profit target with a forty-percent single-day consistency rule. Every design decision in the system reflects these constraints.

The system operates exclusively during the New York morning session from 9:30 AM to 11:30 AM Eastern Time. This window captures the highest liquidity, the strongest institutional intent, and the cleanest price structure of the trading day. No signals are taken outside this window. The selectivity this produces is a feature, not a limitation: a single well-timed trade per week with a high win rate is more valuable to a funded account evaluation than frequent marginal trades that introduce drawdown variance.

The primary research contribution of this work is the empirical validation of a layered filter architecture that combines fixed rule-based logic (Asia range sweep, MSS confirmation) with adaptive, data-driven thresholds (bot memory win rates, VIX regime adjustments, day-of-week profiles) and real-time cross-instrument confirmation (ES/NQ SMT divergence). The interaction between these layers is what produces the high selectivity and correspondingly high win rate observed in backtesting.

2. Theoretical Foundation: Institutional Liquidity Theory

The foundational premise of this system is derived from the work of ICT (Inner Circle Trader), a methodology developed by Michael J. Huddleston that attempts to model institutional order flow in currency and index futures markets. The core insight is that large institutional participants, including market makers, central bank trading desks, hedge funds, and proprietary trading firms, must fill large orders in instruments with limited liquidity. To do so without moving the market against themselves, they require the presence of liquidity on the opposite side of their intended trade.

Liquidity in this context refers to resting limit orders and stop-loss orders that have accumulated at predictable price levels. Retail traders are highly predictable in their stop placement: they place buy stops just above obvious highs, sell stops just below obvious lows, and cluster their orders at round numbers, session extremes, and visible technical levels. This predictability is not a bug in the institutional model; it is a feature. Institutions actively engineer price movements that sweep through these predictable stop clusters, filling their large institutional orders in the process.

2.1 Stop Hunts and Liquidity Sweeps

A liquidity sweep occurs when price extends beyond a well-defined level such as a session high or low, triggers the stop-loss orders clustered there, and then reverses sharply. The extension beyond the level represents the institutional order being filled against the retail stop flow. The reversal represents the institutional position now moving in the intended direction. From the retail trader's perspective, the stop was hit by what appeared to be a breakout that immediately failed. From the institutional perspective, the price was engineered to reach a level where sufficient opposing liquidity existed.

The Asia Session Sweep System is designed entirely around this dynamic. The overnight Asia session (8 PM to midnight Eastern Time) forms a consolidation range with well-defined high and low levels. Every participant who entered a position during the Asia session has stop-loss orders resting just beyond these levels. When New York opens and institutional participants wish to establish large directional positions, one of the most efficient methods available to them is to push price through the Asia range boundary, collect the stops clustered there, and then reverse into the intended direction. The system detects the reversal and enters in the institutional direction once structural confirmation is provided by the Market Structure Shift.

2.2 Why the Asia Range Specifically

The Asia range is used as the primary liquidity reference for several reasons. First, the overnight session produces a well-defined, relatively tight consolidation range in equity index futures because Asian market hours overlap with low-volume US futures trading. The range is visible, distinct, and well-known to market participants, which means stop orders cluster there reliably. Second, the transition from the overnight session to the New York open is the highest volume event of the trading day in equity index futures, providing the institutional order flow necessary to engineer a sweep and subsequent reversal. Third, the Asia range levels serve as a daily reset: each morning provides a fresh, independent setup rather than requiring the trader to carry context from multiple days.

It is important to distinguish between a liquidity sweep and a breakout. A true breakout occurs when price extends beyond a level and continues in that direction with increasing volume and momentum. A liquidity

sweep occurs when price briefly extends beyond a level, triggers the stops there, and then rapidly reverses. The distinction is confirmed by the Market Structure Shift described in Section 5. Without MSS confirmation, the system does not classify a price extension as a sweep and takes no action.

3. The Three-Session Market Framework

Global equity index futures trade on a twenty-three-hour schedule divided into three overlapping sessions, each with distinct liquidity characteristics and participant profiles. Understanding the behavioral profile of each session is essential to the system's logic.

3.1 Asia Session (8:00 PM to Midnight EST)

The Asia session in the context of US equity index futures corresponds to the period when Asian and early European markets are active but North American participants are largely absent. Volume is low, spreads are relatively wide, and price action tends to be consolidatory rather than directional. This session is where the overnight range forms. The system records every five-minute bar during this window and computes the session high and low. These two levels become the primary liquidity reference for the following New York session.

The width of the Asia range is also informative. A narrow range (under twenty points in NQ) indicates that very little price discovery occurred overnight and that stop orders have accumulated in a tight cluster. Wider ranges indicate more directional intent during the overnight session and may correspond to global macro events. The system tracks the Asia range width and includes it as metadata in the backtest trade record, though it is not currently a scored confluence point.

3.2 London Session (2:00 AM to 8:00 AM EST)

The London session represents the arrival of European institutional participants, the world's largest forex and equity markets by volume. London participants frequently establish positions during their morning hours that predate the New York open by four to six hours. In practice, this means that the London session often sweeps one side of the Asia range before New York traders are awake. Empirically, across the sixty-day backtest period, London swept the Asia Low on the majority of days that produced a valid long signal, and swept the Asia High on the majority of days that produced a valid short signal.

The London sweep direction provides a powerful directional bias filter. When London sweeps the Asia Low, it means that institutional participants have already collected the buy stops below the range and established long positions. The subsequent New York long setup that sweeps the same level, or produces a false breakdown, is trading in alignment with the London institutional position. Conversely, a New York signal that goes against the London direction requires an additional confirmation point and is viewed as a lower-probability setup.

The system detects London sweeps by scanning all bars with timestamps between 2:00 AM and 8:00 AM EST and checking whether any bar's high exceeded the Asia session high or any bar's low fell below the Asia session low. The direction is classified as bullish (swept low), bearish (swept high), both (both sides swept, ambiguous), or neutral (neither side swept). This classification is displayed in the morning briefing and affects the minimum score threshold as described in Section 13.

3.3 New York Session (9:30 AM to 11:30 AM EST)

The New York morning session is the only window where the system takes trades. The 9:30 AM opening represents the arrival of the largest and most liquid equity market in the world. Volume spikes dramatically at the open and remains elevated through the first two hours. Institutional participants execute their largest orders during this window, and the price action during this period reflects the most authentic institutional intent of the trading day.

The session is subdivided into two windows. The prime window (9:30 AM to 11:00 AM) is where the majority of institutional order flow occurs. Signals that fire during this window receive an additional confluence point (point 5 of 9). The secondary window (11:00 AM to 11:30 AM) is still tradeable but the minimum required score is automatically raised because momentum tends to fade and reversals become less reliable as the session ages. No signals are accepted after 11:30 AM regardless of score.

4. ICT Power of Three: Weekly Manipulation Model

One of the most important and least discussed concepts in ICT methodology is the Power of Three, also known as the AMD model: Accumulation, Manipulation, Distribution. This model describes institutional behavior not on a daily basis but on a weekly basis. Each week has a primary directional move that institutions intend to execute. The execution of this move follows a structured pattern across the five trading days.

4.1 Weekly Phase Descriptions

Monday (Accumulation): Institutions are establishing their initial weekly positions. Price action tends to be choppy and range-bound because neither side has committed fully to direction. Both the Asia High and Asia Low may be swept on Monday as institutions build positions from both sides. Monday is historically the best trading day in the sixty-day backtest with a 100 percent win rate across three qualifying trades, likely because the clean sweep-and-reverse pattern is most pure when institutional accumulation is occurring.

Tuesday (Manipulation / Judas Swing): This is the most dangerous day for the Asia Sweep System. Tuesday is the day when institutions manufacture a move in the wrong direction specifically to collect stops and create the illusion of a trend before reversing hard on Wednesday. The ICT term for this is the Judas Swing, named after the biblical figure who led followers toward a false destination. On Tuesday, NQ will frequently sweep a level, produce what appears to be a legitimate MSS, and then fail completely. The system recorded zero qualifying trades on Tuesday in the final backtest configuration, because the 8-out-of-9 threshold required on that day is almost never reached by the setups that form.

Wednesday (Distribution): The real weekly direction begins. Institutions have collected enough liquidity and established enough of their position that they can now move the market in the intended direction. Wednesday setups in the backtest had a 100 percent win rate across four qualifying trades. The sweep-and-reverse pattern is at its cleanest on Wednesday because institutional order flow is most aligned with the signal direction.

Thursday (Continuation): The jobless claims release at 8:30 AM EST every single Thursday introduces a binary news spike before the trading session. This spike is unpredictable in direction and magnitude and destroys the clean sweep structure the system relies on. A hard blackout from 8:10 AM to 8:45 AM is enforced. Beyond the blackout, the entire Thursday session carries an elevated minimum score requirement of 8 out of 9. In the final backtest, zero qualifying trades were taken on Thursday.

Friday (Close): Institutions are closing weekly positions and taking profits. Reversals are common near the end of the week as hedges are unwound. Friday follows normal rules in the system with a minimum score of 4 out of 9. The Friday win rate in the backtest was 50 percent across two qualifying trades, consistent with a day that has neither the clean institutional intent of Monday/Wednesday nor the structural trap of Tuesday.

Day	ICT Phase	Institutional Behavior	Min Score	Backtest WR
Monday	Accumulation	Range building, stop collection both sides	4/9	100% (3/3)
Tuesday	Manipulation	Judas Swing, fake direction before real move	8/9	0 trades taken
Wednesday	Distribution	Real weekly direction executes cleanly	4/9	100% (4/4)
Thursday	Continuation	Claims spike 8:30 AM, noise all session	8/9	0 trades taken

Friday	Close	Profit taking, partial reversals	4/9	50% (1/2)
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5. Market Structure Shifts and Displacement Theory

The Market Structure Shift (MSS) is the most critical confirmation signal in the system. A sweep of the Asia range without an MSS is not a trade signal; it is merely a note that liquidity was collected. The MSS provides the structural evidence that the sweep was institutional in nature and that the reversal is underway.

5.1 Definition of MSS

A bullish MSS occurs when, following a sweep below the Asia Low, price closes above a prior swing high established after the Asia session opened. This close above the prior swing high represents a structural break in the prevailing short-term downward price structure. It means that the move below the Asia Low was a trap, and that buyers have taken control sufficiently to displace price above a level that had previously acted as resistance. A bearish MSS is the mirror image: following a sweep above the Asia High, price closes below a prior swing low.

The swing pivots used as MSS reference points are computed using a pivot strength parameter that defaults to 2, meaning a local high requires 2 bars of lower highs on both sides to be classified as a swing high. This prevents the system from triggering on micro-pivots that are not structurally meaningful. The parameter is configurable in both the Python bot and the TradingView Pine Script.

5.2 Displacement Strength Measurement

Not all MSS candles are equal. A candle that closes fractionally above a swing high with a small body and long wicks is technically an MSS but carries little institutional conviction. The system evaluates every MSS candle for displacement strength using two metrics:

- **Body ratio:** the ratio of the candle body (absolute difference between open and close) to the total range (high minus low). A ratio above 0.55 indicates that the majority of the candle's range was covered by a single-direction move with minimal counter-pressure.
- **Relative size:** the ratio of the current candle's range to the 10-bar average range. A ratio above 1.3 means the candle is meaningfully larger than recent candles, indicating an abnormal influx of directional order flow.

When both conditions are met, the MSS is classified as strong. A strong MSS does not add a scoring point in the nine-point system, but a weak MSS (either condition failing) raises the minimum required score by one point. This asymmetric treatment reflects the empirical observation that weak MSS confirmation is a leading indicator of subsequent signal failure.

In the Pine Script visualization, strong MSS candles are marked with a bright green or red label reading 'MSS UP STRONG' or 'MSS DOWN STRONG'. Weak MSS candles are labeled with reduced opacity. The score table in the top-right corner of the chart shows the current MSS strength status at all times.

6. Fair Value Gaps and Order Block Theory

6.1 Fair Value Gaps

A Fair Value Gap (FVG) is an imbalance in price created when a candle's range does not overlap with the ranges of the candles two bars before it. Specifically, for a bullish FVG: the high of the candle two bars ago is lower than the low of the current candle, leaving a gap in the price record where no two-directional trading occurred. For a bearish FVG: the low of the candle two bars ago is higher than the high of the current candle.

FVGs are significant in ICT theory because they represent areas where price moved so rapidly that the market did not have time to find equilibrium. Price has a tendency to return to these zones to 'fill the gap' before continuing in the original direction. When a FVG is present in the entry area at the time of MSS confirmation, it indicates that (a) the displacement was sufficiently strong to create an imbalance, and (b) that there is a structural reason for price to return to the entry area before continuing, making the limit order entry more likely to fill.

The system requires a minimum FVG size of 2 points to filter out microstructure noise. When an FVG is present and price is within its range at signal time, point 3 of 9 is awarded. FVG zones are shown as translucent green (bullish) or red (bearish) boxes on the Pine Script chart, extending 20 bars to the right of their formation.

6.2 Order Blocks

An order block is the last candle in a series that moves counter to the subsequent displacement. In the context of a bullish setup: after the sweep and MSS confirmation, the system scans backwards from the MSS bar to find the last bearish candle (close below open) before the bullish displacement began. This candle represents the bar where institutional sell orders were previously placed and, by implication, where institutional buy orders are now resting. The order block becomes the limit entry target.

The order block entry has two practical advantages over a fixed-offset entry. First, it places the entry at a price where institutional participants are likely to have buy interest, reducing the probability of continued adverse movement. Second, it typically produces a tighter stop loss because the stop is placed one point beyond the order block candle's wick rather than beyond the full sweep extreme. Tighter stops allow a larger reward-to-risk ratio for the same dollar risk.

The system requires a minimum order block body size of 3 points to ensure the block represents meaningful institutional activity rather than a small doji or indecision candle. If no valid order block is found within 20 bars of the MSS confirmation, the system falls back to a fixed 25-point stop entry referenced from the sweep extreme.

7. The Nine-Point Confluence Scoring System

The confluence scoring system is the gatekeeper of the strategy. Each potential signal is evaluated against nine binary conditions. Each condition that is met contributes one point to the score. The minimum required score to fire a signal is four points by default, but this floor is raised dynamically by the smart filter based on market context, day of week, streak state, and session time. Points 1 and 2 are required for any trade to be considered. The remaining seven points differentiate marginal setups from high-conviction entries.

Point	Condition	Technical Basis	Required?
1	Asia sweep detected	Price extended beyond session H/L by 20-80 pts	Yes
2	MSS confirmed	Price closed beyond a prior swing point after sweep	Yes
3	FVG present	Imbalance zone exists in entry area (min 2 pts)	No
4	VWAP aligned	Price on correct side of daily VWAP at signal	No
5	Prime time window	Signal fires between 9:30 AM and 11:00 AM EST	No
6	PDH/PDL confluence	Sweep also took out prior day H or L level	No
7	Opening range opposed	Setup direction reverses the 9:30 AM open move	No
8	Weekly level confluence	Sweep also took out prior week H or L level	No
9	OTE zone entry	Price in 61.8-78.6% Fibonacci retracement zone	No

Two additional quality indicators affect the threshold but do not add points: London alignment (does the London sweep direction match the signal direction?) and MSS displacement strength (is the MSS candle institutionally sized?). Each quality factor that fails raises the minimum required score by one point, meaning a signal with both weak MSS and opposing London direction requires a score of six or higher to fire under base conditions.

7.1 Score Distribution in Backtest

In the final 60-day backtest, scores were distributed as follows across the nine qualifying trades: two trades scored 5/9 (win rate 100%), four trades scored 6/9 (win rate 75%), two trades scored 7/9 (win rate 100%), and one trade scored 8/9 (win rate 100%). The single loss in the backtest occurred at a score of 6/9 on a Friday. There were no trades at 9/9 in the 60-day window, indicating that the highest-confluence setups are rare but highly reliable. The correlation between score and win rate is monotonically positive, supporting the hypothesis that the scoring system correctly measures signal quality.

8. Optimal Trade Entry: Fibonacci Retracement Theory

The Optimal Trade Entry (OTE) concept in ICT methodology refers to the price zone where the best risk-adjusted entry occurs after a sweep and displacement. The theory holds that after an institutional sweep collects liquidity and price begins to reverse, there is frequently a retracement back toward the sweep extreme before the full reversal continues. This retracement provides an entry at a price closer to the sweep low (for longs) or sweep high (for shorts), tightening the stop and improving the reward-to-risk ratio relative to entering at the MSS confirmation bar.

8.1 OTE Zone Computation

The OTE zone is computed using the classic Fibonacci retracement between two structural points. For a long setup:

- Swing origin: the highest high in the 30 bars immediately preceding the sweep bar. This represents the last significant high before the institutional move down that created the sweep.
- Sweep extreme: the actual low of the sweep bar, the lowest price reached during the liquidity collection.
- OTE zone: the range from 61.8 percent to 78.6 percent of the distance from the sweep extreme back toward the swing origin. Expressed as absolute prices: OTE Low = sweep extreme + (swing origin minus sweep extreme) times 0.618. OTE High = sweep extreme + (swing origin minus sweep extreme) times 0.786.

For a short setup, the logic is inverted. The swing origin is the lowest low in the 30 bars before the sweep, the sweep extreme is the high of the sweep bar, and the OTE zone is computed as a retracement downward: OTE High = sweep extreme minus range times 0.618, OTE Low = sweep extreme minus range times 0.786.

8.2 Why 61.8 and 78.6 Percent

The 61.8 percent level is the classic Fibonacci golden ratio retracement and is one of the most widely recognized levels in technical analysis. The 78.6 percent level is the square root of 0.618 and represents the deepest retracement that can occur before a move is considered to have failed. A retracement beyond 78.6 percent of the prior swing in the correct direction suggests the original move has been fully corrected. Entries within the 61.8 to 78.6 percent zone place the stop just beyond the 100 percent retracement level, providing a structurally defined invalidation point.

In practice, the OTE zone provides a limit entry target that is more favorable than the MSS confirmation price. Not all setups will have price retrace into the OTE zone before continuing; some will simply accelerate from the MSS bar. When OTE alignment does occur, point 9 of 9 is scored. The backtest showed OTE entries at 100 percent win rate across the limited sample of 1 qualifying OTE trade, consistent with theory but not yet statistically significant. OTE is visualized as a semi-transparent box on the Pine Script chart.

9. SMT Divergence: ES/NQ Confirmation Analysis

Smart Money Technique (SMT) divergence is a cross-instrument confirmation methodology that compares the price behavior of two correlated instruments at structurally significant levels. In the context of equity index futures, NQ (Nasdaq-100) and ES (S&P; 500) are highly correlated instruments that generally sweep key levels in synchrony when the move is institutionally driven.

9.1 The SMT Divergence Signal

SMT divergence occurs when one instrument sweeps a structural level while the correlated instrument does not. Specifically, in a long setup context: NQ sweeps below its Asia Low but ES does not make a corresponding new low below its own Asia Low. This divergence indicates that the NQ move was not broadly supported by the equity market as a whole. Institutions use ES as their primary execution instrument because of its superior liquidity and tighter bid-ask spreads. When NQ makes a new low but ES does not, it suggests the NQ move was manufactured specifically to collect NQ-specific stop orders, without genuine broad market weakness behind it.

The ICT interpretation of SMT divergence is as follows: if NQ sweeps its Asia Low and ES simultaneously sweeps its own Asia Low, both instruments are moving together and the setup reflects genuine broad market institutional activity. If NQ sweeps its Asia Low but ES holds above its Asia Low, the NQ move is divergent from the broader market, indicating a weaker institutional conviction behind the setup. In the context of the Judas Swing model, Tuesday SMT divergences represent the fake weekly move: NQ is engineered lower to collect stops while ES remains relatively stable, confirming the move is manufactured.

9.2 Implementation

At the moment a sweep is detected, the system downloads ES=F five-minute bar data and identifies the closest ES bar to the NQ sweep timestamp. It then checks whether the ES bar at that time also swept the corresponding ES Asia range boundary. If ES did not sweep, the `smt_confirmed` flag is set to False and the signal is hard-blocked regardless of score.

Backtest result: the single SMT-divergent trade in the 60-day dataset had a P&L; of negative forty-one dollars (a loss). The win rate for SMT-divergent trades was zero percent. This is a very small sample but strongly consistent with the theoretical prediction. The hard block was implemented after this observation.

10. VIX Regime Classification and Volatility-Adjusted Thresholds

The CBOE Volatility Index (VIX) measures the market's expectation of 30-day implied volatility in the S&P; 500, derived from options prices. It is widely used as a gauge of market fear and uncertainty. Different VIX regimes create fundamentally different trading environments, and a strategy designed for low-volatility conditions may perform poorly during extreme volatility and vice versa.

10.1 VIX Regime Classification

The system classifies VIX into five regimes at startup by downloading the most recent VIX daily close from Yahoo Finance:

VIX Level	Regime	Market Interpretation	Score Penalty
Below 15	Low	Complacency, trend following dominates	+0
15 to 20	Medium	Normal conditions	+0
20 to 25	High	Elevated uncertainty, trends volatile	+0
25 to 30	Very High	Fear elevated, sharp intraday moves	+0
Above 30	Extreme	Crisis regime, unpredictable spikes	+1

10.2 Empirical Finding: VIX Penalty Inverted for Short Setups

A counterintuitive finding from the backtest was that high VIX environments (above 20) actually produced a 100 percent win rate in the 60-day dataset across five qualifying trades. This is the opposite of the initial assumption that high volatility would hurt the reversal model. The explanation is directional: the 60-day window coincided with a strongly bearish NQ environment with elevated VIX. In a trending bearish market, short sweep setups are aligned with the broader institutional trend. High VIX during a downtrend means institutional participants are aggressively positioned short, making the short sweep reversals stronger and more reliable.

As a result, the original VIX score penalty was removed from the minimum score calculation for this implementation. Only extreme VIX (above 30) carries a penalty of one additional point. This reflects the recognition that VIX alone does not determine whether a sweep reversal will succeed; the direction of the VIX regime relative to the signal direction is the relevant variable. A more sophisticated future implementation might apply the VIX penalty selectively to long signals during high VIX environments (since rising VIX in a downtrend suggests trend continuation rather than reversal) while leaving short signals unpenalized.

11. Economic Calendar Integration and News Blackout Protocols

High-impact economic news releases create rapid, binary, unpredictable price movements that are fundamentally incompatible with the sweep-and-reversal model. The Asia Sweep System depends on institutional order flow creating a clean sweep of a defined level followed by a structural reversal. A news spike can create a sweep of identical technical appearance that is driven entirely by algorithmic news reaction rather than institutional liquidity collection. The system addresses this risk through hardcoded calendar-based blackout windows.

11.1 Thursday Jobless Claims

The most impactful recurring news event for NQ futures is the Initial Jobless Claims report released by the US Department of Labor every Thursday at 8:30 AM Eastern Time. This report consistently produces price spikes of 50 to 150 NQ points within seconds of release. The spike direction is unpredictable: a lower-than-expected claims number is bullish, a higher number is bearish, and the market's reaction often reverses the initial spike direction within minutes as the data is digested.

The system enforces a hard blackout from 8:10 AM to 8:45 AM every Thursday. Any sweep or MSS detected within this window is discarded. Beyond the blackout window, the Thursday session carries a +1 score penalty for the full day, raising the minimum required score from 4 to at least 5. Combined with the base Thursday threshold of 8 out of 9 (set based on backtest analysis), this effectively blocks all but the highest-conviction Thursday setups.

11.2 Monthly High-Impact Events

The 2026 dates for the following events are hardcoded into the system's economic calendar. Each event carries a 20-minute pre-release blackout and a 15-minute post-release blackout for the release window, plus a +2 score penalty for the full session:

- Non-Farm Payrolls (NFP): First Friday of every month, 8:30 AM EST
- Consumer Price Index (CPI): Monthly, variable date, 8:30 AM EST
- Federal Open Market Committee (FOMC) Rate Decisions: Eight times per year, 2:00 PM EST

FOMC days are particularly dangerous because the market trades in an information vacuum for the hours leading up to the 2:00 PM decision, creating false technical setups that resolve violently at the announcement. The +2 session-wide score penalty on FOMC days makes it extremely unlikely that the system will fire a signal, which is the intended behavior. The morning briefing displays the news calendar status prominently so the trader is aware of the elevated-risk environment before the session begins.

12. The Four-Layer Market Context System

The four-layer market context system is the intelligence backbone of the bot, assembled fresh each morning and updated dynamically when sweeps are detected. It synthesizes external market data from three sources (VIX, ES futures, economic calendar) and derived data from the bar history (weekly levels) into a unified context dictionary that downstream components query when making threshold decisions.

12.1 Architecture

The context builder is implemented in `strategy/market_context.py`. It exposes the following key functions:

- `get_vix_regime()`: Downloads VIX daily data, classifies regime, returns penalty value
- `get_es_data()`: Downloads ES=F five-minute bars for the backtest period
- `check_smt_divergence()`: Compares NQ and ES sweep behavior at a given bar
- `compute_ote_zone()`: Computes 61.8-78.6 percent Fibonacci zone from swing data
- `check_news_calendar()`: Returns skip flag and score penalty for current timestamp
- `get_weekly_levels()`: Computes prior week H/L from the bar dataframe
- `get_weekly_profile()`: Returns the ICT day phase and guidance for today
- `context_score_penalty()`: Master function returning (skip, penalty, reason) tuple

12.2 Context Dictionary Structure

The assembled context dictionary passed to the smart filter and scorer contains: `vix` (regime and penalty), `weekly` (`prev_week_high` and `prev_week_low`), `profile` (ICT phase, guidance, quality note), `news` (skip flag, score penalty, reason), `smt` (confirmed, divergent, detail string), `ote` (`ote_high`, `ote_low`, valid flag), `trade_date`, and `day_of_week`. Each downstream component extracts the fields it needs. The smart filter reads `news`, `vix`, and `smt`. The confluence scorer reads `weekly` for point 8 computation and receives `ote_zone` as a direct boolean parameter.

13. The Smart Adaptive Filter Architecture

The SmartFilter class in strategy/smart_filter.py is the dynamic threshold manager. It receives the current bar state and market context and returns the minimum score required to fire a signal. Its design is deliberately layered: each successive layer can only raise the threshold, never lower it below what a prior layer set, except for the Monday fast-pass which is explicitly allowed to lower it in highly specific conditions.

13.1 Layer 1: Day-of-Week Base Threshold

The base threshold is the starting point before any adjustments. If the bot has accumulated enough trade data to use the BotMemory module, it uses the observed win rate for the current day of week to set the threshold. If the win rate is below 30 percent, the base is 7 out of 9. If below 45 percent, the base is 5. Otherwise the base is 4. Without sufficient memory data, the hardcoded baselines apply: Monday and Wednesday at 4, Friday at 4, Tuesday at 8, Thursday at 8.

13.2 Layer 2: Market Context Penalty

After the DOW base is set, market context penalties are applied. The news skip flag (True during blackout windows) returns 99 immediately, which is treated as a skip signal by the engine. News score penalties add to the base. SMT divergence adds 1 to the base. VIX penalties add to the base only in extreme regimes. These additions use max() with the current base to ensure the threshold never decreases.

13.3 Layer 3: Session-State Rules

Session-state rules apply on top of the context-adjusted base:

- After two or more consecutive losses: base raised to at least 6
- After 11:00 AM EST: base raised to at least 5
- Weak MSS candle: base raised by 1
- London direction opposed to signal: base raised by 1
- Sweep depth below 15 points: base raised to at least 6

13.4 Layer 4: Monday Prime-Window Fast-Pass

The only mechanism that can lower the threshold is the Monday fast-pass. On Mondays before 10:00 AM, if the bot is on a win streak of at least one consecutive win, the threshold is allowed to drop to a minimum of 4. This is based on the empirical observation that Monday is the strongest day historically and that a winning bot in a favorable environment should be willing to take slightly lower-confidence entries during the prime window.

13.5 Absolute Cap

The smart filter caps its output at 8. This means that no combination of penalties can require a score higher than 8 out of 9. A score of 9 would require all nine conditions simultaneously, which is virtually impossible in practice. Capping at 8 ensures the system can still fire when conditions are genuinely excellent even during elevated-risk environments.

14. Risk Management and Prop Firm Compliance

The risk management framework is designed to satisfy two objectives simultaneously: protecting the account from catastrophic drawdown and maximizing the probability of passing the Tradeify evaluation within the 60-day window. These objectives are sometimes in tension: more conservative filters produce fewer trades but higher win rates, while looser filters produce more trades with higher drawdown variance.

14.1 Per-Trade Risk

Each trade risks a maximum of fifty dollars, corresponding to a 25-point stop on 1 MNQ contract at two dollars per point. This represents 0.2 percent of the 25,000-dollar starting balance. The position sizer computes the number of contracts such that the dollar risk never exceeds `MAX_RISK_PER_TRADE` regardless of the stop distance. For a 20-point stop, the risk is forty dollars per contract. For a 30-point stop, fifty dollars caps the position at 1 contract.

14.2 Trailing Maximum Drawdown

The Tradeify trailing maximum drawdown rule states that the account balance may never fall more than one thousand dollars below the highest balance ever reached. This is a trailing drawdown, meaning it follows the peak balance upward but never adjusts downward. The `account_state` module tracks the peak balance and computes the current drawdown floor at all times. If the balance approaches within two hundred dollars of the floor, the morning briefing flags this in red. If the floor is breached, all signal generation stops.

14.3 Daily Loss Limit

A session-level daily loss limit of one hundred dollars (two maximum-risk trades) is enforced independently of the trailing drawdown calculation. After two consecutive maximum-risk losses in a single session, the bot stops generating signals for that day. This prevents a bad session from causing disproportionate damage to the evaluation's drawdown buffer.

14.4 Consistency Rule

The Tradeify consistency rule states that no single day's profit may exceed 40 percent of total cumulative profit. The bot enforces this at 38 percent, providing a 2 percent safety margin. Once cumulative profit is positive, the bot computes the maximum profit allowed today (38 percent of total profit) and auto-stops signal generation when that amount is approached. This is displayed in the morning briefing's Consistency Cap panel.

14.5 Maximum Trades Per Day

The system limits itself to a maximum of two trades per day. This is a behavioral discipline constraint: after two setups, the morning session's primary setups have been addressed and any additional trades are likely to be marginal or revenge-driven. In the 60-day backtest, the system never took more than two trades on any single day due to the strict filtering.

15. Persistent Bot Memory and Adaptive Threshold Calibration

The BotMemory module in `bot_memory.py` implements a simple but effective learning mechanism that allows the system's threshold baselines to update over time as real trade data accumulates. Unlike the static backtest baselines that are derived from historical data, the bot memory reflects the system's actual live performance in the specific account and market conditions it is being run in.

15.1 Data Collected Per Trade

Each completed trade records outcome data into the following categorical buckets: day of week, confluence score, signal hour (9, 10, or 11), London alignment status, MSS strength status, sweep depth bucket (shallow, normal, deep), entry type (order block or fixed), and direction (long or short). These buckets allow the system to develop independent win rate estimates for each combination of conditions.

15.2 Minimum Sample Requirement

The system requires a minimum of eight trades in a given bucket before the observed win rate is trusted over the hardcoded baseline. With fewer than eight trades, the baseline applies. This prevents the system from over-adjusting its behavior based on small sample statistical noise. Eight trades is a conservative threshold; a more sophisticated Bayesian approach using a beta distribution prior would provide smoother updating with smaller samples, and is a candidate for future implementation.

15.3 Threshold Recalculation Logic

After each trade is recorded, the `recalculate_thresholds` function runs across all populated buckets. The rule is: if the observed win rate in a bucket is below 30 percent, the minimum score for that category is set to 7 out of 9. If below 45 percent, it is set to 5. Otherwise, the default minimum of 4 applies. These thresholds were chosen to be conservative: a 45 percent win rate with a 3:1 reward-to-risk ratio is slightly positive in expectation, but below the system's target performance level.

The memory file is written to `memory/bot_memory.json` after every trade and is read at startup. The briefing system displays a note when the bot is operating from live data versus baseline data for any given category, so the trader knows when the system has accumulated sufficient experience in that dimension.

16. Backtest Methodology

The backtest engine in `backtest/engine.py` replays historical five-minute NQ futures bars through the complete strategy pipeline, simulating the same detection logic, scoring, filtering, and trade management that the live bot uses. The backtest is designed to be as realistic as possible while acknowledging the inherent limitations of bar-level simulation.

16.1 Data Source

All historical price data is sourced from Yahoo Finance via the `yfinance` Python library using the ticker `NQ=F` (continuous front-month E-mini Nasdaq-100 futures). The default period is 60 calendar days at 5-minute intervals, producing approximately 13,000 to 14,000 bars per run. ES futures data (`ES=F`) is downloaded for SMT divergence checks. VIX data (`^VIX`) is downloaded at daily granularity for the same period.

16.2 Simulation Assumptions

The backtest makes the following simplifying assumptions, each of which could introduce optimistic bias relative to live trading:

- Limit orders fill at the exact limit price with no slippage. In live trading, limit orders at the order block price may not fill if price reaches the level and reverses without touching the exact price, or may fill at a worse price during fast markets.
- Stop losses fill at the stop price. In live markets, gaps or fast price action can cause stop fills at worse prices, particularly around news events.
- No bid-ask spread is modeled. MNQ spreads are typically 0.25 points (\$0.50), which would reduce each trade's P&L; by \$0.50 in the real world.
- The starting balance is set at \$24,600 (the \$25,000 starting balance minus the \$400 already drawn down in the evaluation).

16.3 Trade Simulation Logic

Each signal causes a simulated trade to be entered at the limit entry price. If price never touches the limit entry within the session window, no trade is recorded (limit order not filled). Once filled, the simulation tracks the subsequent bars for stop and TP levels. With one contract: hitting TP1 moves the stop to break-even; hitting TP2 from there is a full win. If the stop is hit at break-even after TP1, the outcome is `WIN_TP1_BE` with zero P&L; If the stop is hit before TP1, the outcome is `LOSS`. End of session closes at the last bar's close price, recorded as `TIMEOUT`.

17. Backtest Results and Filter Iteration Analysis

The backtest was run three times over the same 60-day dataset (March 13 to May 22, 2026) with progressively improved filter settings. The iteration process was data-driven: after each run, the results analyzer identified the worst-performing conditions, and targeted rule changes were made to address them.

17.1 Iteration Summary

Round	Key Change	Trades	Win Rate	P&L	Max DD
Baseline	Original filters (8 pt sweep min)	20	40%	\$803	\$250
Round 2	Tue/Thu threshold raised to 8/9	12	67%	\$1,077	\$250
Round 3	Sweep depth 20 pt min, SMT hard block	9	89%	\$1,118	\$250

17.2 Round 1 Analysis

The baseline run produced 20 trades with a 40 percent win rate. The most significant finding was that shallow sweeps (below 15 points) had a 0 percent win rate across four trades. Tuesday and Thursday combined produced 6 trades with only 1 win, consistent with the Judas Swing and jobless claims theories. The `is_sweep_valid` function existed in the codebase but was not being called in the backtest engine, meaning shallow sweeps were not being filtered at the detection stage.

17.3 Round 2 Analysis

Raising the Tuesday and Thursday thresholds to 8 out of 9 eliminated all 8 Tuesday and Thursday trades. The win rate rose from 40 to 67 percent with 12 remaining trades. The P&L increased from \$803 to \$1,077 despite having fewer trades. The most important finding in Round 2 was the sweep depth distribution: sweeps of 30 to 80 points produced an 89 percent win rate, while sweeps below 30 points produced a 0 percent win rate. This was the primary basis for the Round 3 change.

17.4 Round 3 Analysis

Raising the minimum sweep depth from 8 to 20 points (implemented in the `is_sweep_valid` function and now called at sweep detection) removed 3 additional low-quality trades. Hard blocking SMT-divergent signals removed the single SMT-divergent trade, which had been a loss. The final result was 9 trades at 89 percent win rate and \$1,118 total P&L, with all 9 trades falling in the 30 to 80-point deep sweep category.

17.5 Final Backtest Statistics

Metric	Value
Backtest period	March 13 to May 22, 2026 (60 days)
Trading days in period	49
Total qualifying trades	9
Wins	8

Losses	1
Win rate	88.9%
Average win	\$146.06
Average loss	\$50.00
Profit factor	23.4 (total wins / total losses)
Total P&L	\$1,118.50
Maximum simulated drawdown	\$250.00
Drawdown limit	\$1,000.00
Consistency violations	2
Would pass evaluation	No (needs \$1,500 target)

The system did not pass the evaluation in backtest due to the \$1,500 profit target. At the current rate of approximately 1.5 qualifying trades per week and \$146 average win at 89 percent win rate, the expected weekly P&L; is approximately \$185. The path from the current negative \$400 position to the \$1,500 target is approximately 10 weeks. This is within the typical 60-day evaluation window if the system begins live trading immediately.

18. Day-of-Week Statistical Analysis

The day-of-week breakdown is one of the most informative dimensions of the backtest analysis because it validates the ICT Power of Three model empirically. The pattern is remarkably consistent with theoretical predictions.

Day	Trades	Wins	Losses	Win Rate	Avg P&L	Notes
Monday	3	3	0	100%	+\$146	Accumulation phase. Clean signals.
Tuesday	0	-	-	Blocked	-	8/9 threshold blocked all setups.
Wednesday	4	4	0	100%	+\$150	Distribution phase. Best quality.
Thursday	0	-	-	Blocked	-	8/9 + jobless claims blocked all.
Friday	2	1	1	50%	+\$40	Normal rules. Mixed results.

The Monday and Wednesday results are particularly striking. All seven trades on these two days were wins. Monday's 100 percent win rate is consistent with the Accumulation phase theory: institutional participants are establishing positions in a clean, predictable manner with well-defined sweep structures. Wednesday's 100 percent win rate is consistent with the Distribution theory: the real weekly directional move is underway and sweep signals align precisely with institutional intent.

The complete absence of qualifying trades on Tuesday and Thursday following the threshold increase to 8 out of 9 is a double-edged result. On one hand, it perfectly validates the theory: these days are genuinely problematic and the elevated threshold correctly identifies that no setup reaches near-perfect quality on those days. On the other hand, it means the bot has zero exposure on 40 percent of trading days. Over a sufficiently large sample, occasional high-quality Tuesday or Thursday setups may emerge that the current threshold would correctly permit.

The Friday result (50 percent, one win and one loss across two trades) is inconclusive given the small sample size. Friday follows normal rules with a 4 out of 9 threshold. The single Friday loss was a deep sweep (64 points) at score 6 in medium VIX, with no obvious distinguishing feature that would justify a rule change. It is characterized as normal win-rate variance.

19. Sweep Depth Analysis

Sweep depth analysis was the most actionable finding of the backtest iteration process. The relationship between sweep depth and win rate showed a sharp threshold effect rather than a gradual correlation.

Sweep Depth	Trades (R1)	Win Rate (R1)	Trades (R3)	Win Rate (R3)	Interpretation
0-8 pts	3	0%	0	Rejected	Noise, below inst. threshold
8-15 pts	1	0%	0	Rejected	Shallow, retail noise
15-30 pts	2	0%	0	Rejected	Insufficient displacement
30-80 pts	8	87%	9	89%	Institutional displacement
80+ pts	0	N/A	0	Rejected	News event spike

The pattern is unambiguous: every single winning trade in the dataset involved a sweep depth of 30 points or more. Every losing or break-even trade involved a sweep depth below 30 points (excluding the one deep-sweep Friday loss, which is attributed to normal variance). The minimum sweep depth was set at 20 points in the implementation rather than 30 to provide a small buffer: setting it at exactly the empirical threshold would risk overfitting to the 60-day sample. The 20-point minimum still filters all the observed weak sweeps while allowing for the possibility that occasionally a 20 to 30-point sweep might succeed under optimal broader conditions.

The theoretical explanation for this pattern is consistent with the institutional liquidity model. A shallow sweep (below 20 points) does not penetrate far enough beyond the session level to collect the stop clusters that justify an institutional reversal. These shallow moves are more likely to be retail momentum trading or algorithmic probes rather than deliberate institutional liquidity collection. Deep sweeps of 30 points or more represent a committed institutional move, absorbing all the available liquidity at and beyond the level and creating the preconditions for a clean, sustained reversal.

20. VIX, SMT, and OTE Empirical Results

20.1 VIX Regime Results

VIX Regime	Trades	Win Rate	Avg P&L	Notes
Low (< 15)	0	-	-	No qualifying trades in period
Medium (15-20)	4	75%	+\$95	Three wins, one Friday loss
High (20-25)	4	100%	+\$147	All wins, strongly bearish market
Very High (25-30)	1	100%	+\$150	One win in high-vol period
Extreme (> 30)	0	-	-	No trades in extreme VIX period

The high and very-high VIX trades all occurred during the March-April 2026 market selloff period when NQ was in a sustained downtrend. Short sweep signals during this period were aligned with the macro institutional trend, producing strong continuation moves after each reversal. The medium VIX period (later in the dataset) produced a mix of results, including the single loss. This supports the directional interpretation: VIX level matters less than whether the signal direction is aligned with the VIX-implied institutional trend.

20.2 SMT Divergence Results

Only one SMT-divergent trade occurred in the 60-day dataset. That trade produced a loss of negative forty-one dollars. This is a sample of one and cannot be treated as statistically significant, but it is perfectly consistent with the theoretical prediction: SMT divergence indicates a fake move and produces a losing reversal. The hard block was implemented based on this result and the strong theoretical support.

20.3 OTE Zone Results

The OTE zone check was added in the final iteration. In the 60-day dataset, only one of the nine qualifying trades entered while price was inside the OTE zone, and that trade was a win (\$150 at TP2). Eight trades entered outside the OTE zone: seven wins and one loss. The OTE results are too limited for statistical conclusions, but the theoretical case for OTE entries being higher quality (tighter stop, better R:R) is strong. Over a larger sample, OTE entries are expected to show a meaningfully higher win rate and better average P&L; than non-OTE entries.

21. System Architecture and Pipeline

The system is implemented in Python 3.9+ and consists of approximately 15 modules across four packages: strategy (signal detection), risk (account management), backtest (historical simulation), and a root-level module for live operation. The pipeline from raw price data to trade signal follows a strict sequential architecture:

Stage	Module	Function	Output
1. Data	data_loader.py	yfinance download, session labeling	DataFrame with in_trade_window flag
2. Context	market_context.py	VIX, ES, calendar, weekly levels	Context dict passed downstream
3. London	london_session.py	2-8 AM sweep scan	Direction: bullish/bearish/both/neutral
4. Asia range	asia_range.py	8 PM-midnight H/L	asia_high, asia_low per date
5. Sweep detect	smart_filter.py	is_sweep_valid(), depth check	Boolean + depth value
6. SMT check	market_context.py	ES divergence at sweep bar	smt_confirmed boolean
7. OTE compute	market_context.py	Fibonacci zone from swing data	ote_high, ote_low, valid
8. MSS detect	mss_detector.py	Pivot tracking, displacement	Boolean + is_strong
9. OB find	order_block.py	Last opposing candle scan	entry, stop, stop_distance
10. Score	confluence_scorer.py	9-point boolean check	ConfluenceResult (score, reason)
11. Filter	smart_filter.py	min_score_required(), 4 layers	Min threshold integer or 99
12. News check	market_context.py	Blackout window check	skip boolean
13. Risk check	risk_manager.py	Balance, DD, consistency	approved boolean
14. Signal	live_detector.py	Terminal panel + notification	Entry/stop/TP values
15. Memory	bot_memory.py	Record outcome by category	Updated JSON thresholds

22. Pine Script Visual Reference

The TradingView Pine Script (pine_script/tjr_enhanced.pine) is a visual companion to the Python bot. It is not the trading system itself. Its purpose is to show the trader what the bot is seeing on the chart, to allow manual verification of signals, and to support retrospective analysis of setups. The bot can operate entirely without the Pine Script.

22.1 Visual Elements

Element	Color	Description
Asia H/L lines	Orange	Today's Asia session high and low
PDH/PDL lines	Grey	Previous day high and low
PWH/PWL lines	Purple	Previous week high and low
VWAP	Yellow	Volume-weighted average price
Opening range box	Yellow translucent	9:30 to 10:00 AM range
FVG boxes	Green/red translucent	Fair value gap zones
Order block boxes	Green/red	Last opposing candle before MSS
OTE zone box	Green/red	61.8-78.6% Fibonacci retrace zone
Sweep label	Green up/Red down	Depth and direction of sweep
Shallow label	Grey	Rejected shallow sweep below 20 pts
SMT divergence	Orange	ES did not confirm NQ sweep, blocked
MSS label	Green/Red	Market structure shift, strong or weak
Signal label	Green/Red	Score, entry, stop, TP1, TP2
DOW label	Day-specific	ICT phase name and min score today
Score table	Top-right	Live 9-point breakdown, SMT, London, MSS
Background shading	Blue/grey	Prime window (blue) vs late window (grey)

22.2 Score Table

The 14-row score table in the top-right corner shows: a header row, nine rows for each scoring point (yes/dash for long and short columns), one row for the current score versus required threshold, one row for London direction and opening range direction, one row for MSS strength, and one row for SMT status. This table updates on every bar and gives an at-a-glance view of exactly how many points a potential setup has accumulated.

22.3 Alert Conditions

Seven alert conditions are defined: Long Signal, Short Signal, Asia Low Swept, Asia High Swept, Bullish MSS, Bearish MSS, and SMT Divergence. The Long and Short Signal alerts fire only when all thresholds are met. The SMT Divergence alert fires even when the signal is blocked, notifying the trader that a setup formed but was rejected due to ES/NQ divergence.

23. Configuration Reference

All user-configurable parameters are centralized in config.py. This section documents each parameter, its default value, and its effect on system behavior.

Parameter	Default	Description
STARTING_BALANCE	\$25,000	Account starting balance
TRAILING_MAX_DRAWDOWN	\$1,000	Prop firm trailing drawdown limit
PROFIT_TARGET	\$1,500	Evaluation profit target
CONSISTENCY_RULE	0.40	Single-day profit cap as fraction of total
CONSISTENCY_BUFFER	0.38	Bot-enforced cap (2% below firm rule)
MAX_RISK_PER_TRADE	\$50	Maximum dollar risk per trade
MAX_STOP_POINTS	25	Maximum stop distance in points
MAX_DAILY_LOSS	\$100	Session loss limit before signal pause
MAX_TRADES_PER_DAY	2	Maximum signals per session
MIN_CONFLUENCE_SCORE	4	Floor score (smart filter may raise this)
MNQ_DOLLARS_PER_POINT	2.0	Dollar value per index point
TICKER	NQ=F	yfinance ticker for price data
TRADE_START_HOUR	9	Session start hour (EST)
TRADE_START_MINUTE	30	Session start minute
TRADE_END_HOUR	11	Session end hour
TRADE_END_MINUTE	30	Session end minute

24. Limitations and Known Failure Modes

Any honest assessment of a trading system must include its failure modes. The following limitations are known and accepted aspects of the current implementation.

24.1 Small Backtest Sample

Nine trades over 60 days is a very small sample for statistical validation. The 89 percent win rate is strongly consistent with the theoretical model, but a small sample can mask longer-term win rates. A 90-percent confidence interval around 89 percent with $n=9$ is approximately 52 percent to 100 percent. The true long-run win rate could be significantly lower. Extended backtesting over 12 to 24 months of data would substantially reduce this uncertainty.

24.2 Directional Bias in Backtest Period

The March to May 2026 backtest window coincided with a strongly bearish NQ market. All nine qualifying trades were short. This is directionally appropriate given the market structure, but it means the system has no validated long-side signals in the backtest dataset. Long-side sweep signals in uptrending markets may behave differently and may require different filter calibrations.

24.3 Slippage and Fill Assumptions

The backtest assumes perfect limit order fills at the limit price with no spread. In live trading, MNQ limit orders may experience partial fills during fast markets, fills at the next tick if the limit is never exactly touched, or no fill at all if price reverses before reaching the limit. The real-world P&L of a live system will be modestly lower than backtest results due to these execution frictions.

24.4 yfinance Data Quality

yfinance provides free historical data but with known data quality issues: occasional missing bars, incorrect timestamps around daylight saving time transitions, and potential gaps during extended hours. These issues can occasionally produce incorrect Asia range calculations or missed session boundaries. The system does not currently validate data completeness before running.

24.5 ES Ticker Availability

The SMT divergence check requires access to ES=F data from the same data source as NQ. If ES data is unavailable (network error, market holiday, data vendor issue), the SMT check defaults to 'confirmed' and signals are not blocked. This is a graceful degradation but reduces the quality of the filter on those occasions.

24.6 Thursday and Tuesday Blockout Completeness

The 8 out of 9 threshold on Tuesday and Thursday is so high that it effectively blocks all trading on those days. In theory, there exist setups on those days that score 8 or 9 and would be valid trades. In practice, none appeared in the 60-day window. This may represent a genuine structural absence of qualifying setups on those days, or it may reflect the specific market conditions of the test period. A more nuanced approach might implement a situational Tuesday filter that allows trading after the Judas Swing has reversed (i.e., after the

fake move has already played out), rather than blocking the entire day.

25. Morning Briefing and Operational Workflow

The morning briefing is the first output the system produces each trading day, generated automatically when the bot starts up before the New York session. Its purpose is to give the trader a complete situational picture before any signals fire, so that no decision is made in the heat of a developing setup. Every piece of context that could affect a trade decision is surfaced in the briefing, read once, and then the trader simply watches the bot execute its logic.

25.1 Briefing Panel Contents

The morning briefing prints to the terminal in structured sections. Each section is designed to answer one question the trader might ask before the session begins. The panels appear in the following order:

- **Account Status:** current balance, equity peak, current drawdown, buffer remaining before the trailing drawdown floor is hit, and the daily P&L; required to reach the profit target.
- **Risk Parameters:** the per-trade risk in dollars, the calculated position size in contracts given current stop distance settings, and the consistency cap for today (maximum profit before the 40 percent single-day rule is triggered).
- **Market Context:** VIX regime and today's implied volatility classification, whether ES data loaded successfully for SMT checks, and any pending economic events with their exact blackout windows.
- **Day of Week Profile:** the ICT phase name for today (Accumulation, Manipulation, Distribution, Continuation, or Close), the recommended trading stance, and the minimum confluence score threshold in effect for the day.
- **London Session Summary:** the direction of London's sweep of the Asia range (bullish, bearish, both, or neutral), and whether London direction alignment will provide a confluence bonus or penalty for signals that fire this session.
- **Asia Range:** the exact high and low prices of last night's Asia session, the range width in index points, and the opening range direction from the prior 9:30 to 10:00 AM window if this is not the first run of the day.
- **Memory Status:** how many live trades have been recorded, which DOW buckets have enough data to use live-calibrated thresholds, and any threshold adjustments the memory module has made relative to the hardcoded baselines.

25.2 Pre-Session Checklist Protocol

The recommended protocol before each session is: run the bot, read the briefing, note the minimum required score and any elevated-risk flags, open TradingView with the Pine Script chart at the same timeframe, and confirm that the Asia range lines on the chart match the values the bot printed. This cross-check catches any data discrepancy between yfinance (bot source) and TradingView's data feed before the session begins. If the levels disagree by more than 2 points, the TradingView data should be trusted as the more reliable real-time source and the bot's Asia range should be manually overridden.

25.3 Signal Alert Structure

When a qualifying signal fires, the bot outputs a structured alert block to the terminal and, if configured, sends a notification via the alerts module. The alert block includes: signal type (LONG or SHORT), timestamp, entry price (order block level), stop loss price, TP1 price (1:1 reward), TP2 price (2:1 reward), confluence score and breakdown, minimum score that was required, reason string from each scoring point, SMT status, OTE status, VIX regime at signal time, and the position size in contracts. The breakdown is printed in full so the trader can immediately understand why the signal fired and can override if they observe something the bot cannot detect, such as breaking news or a visible chart pattern that contradicts the signal.

25.4 End-of-Session Summary

After the 11:30 AM session close, the bot prints a session summary: trades taken today (if any), P&L, running balance, updated drawdown status, and a reminder of the consistency cap calculation for tomorrow. The bot then enters a dormant state until 8:00 PM when the next Asia session begins and data collection resumes.

26. Trade Execution and Order Management

This section describes the complete lifecycle of a trade from signal generation to final close, including the order types used, the stop management logic, and the session close protocol. Understanding this lifecycle is essential for anyone who intends to automate execution through the Tradovate API or to manually replicate the bot's order management logic.

26.1 Entry Order Placement

The entry order is a limit order placed at the order block level. The order block level is the midpoint of the last opposing candle before the MSS displacement. The limit order is good for the remainder of the current session. If price does not retrace to the order block level before 11:30 AM, the order is cancelled and no trade is taken. This limit entry approach is preferable to a market entry at the MSS bar because it provides a better price, a tighter stop distance, and confirmation that institutional participants are supporting the level as expected.

When no valid order block is found within the lookback window, the entry falls back to a fixed 5-point retracement from the MSS confirmation bar's close. This fallback entry produces a wider stop distance and lower reward-to-risk ratio, which is reflected in the scoring system: the absence of a clean order block is not a scored point, but it implicitly reduces the quality of the setup by increasing risk.

26.2 Stop Loss Calculation

The stop loss is placed one point beyond the extreme of the sweep bar. For a long setup, the stop is one point below the sweep low. For a short setup, one point above the sweep high. This placement reflects the logical invalidation point: if price returns below the sweep low, the sweep was not reversed, and the institutional long thesis is invalidated. One point of buffer is added to avoid the stop being triggered by the spread on exact fills.

The stop distance is also subject to the MAX_STOP_POINTS cap. If the distance from the order block entry to the sweep extreme exceeds 25 points, the stop is moved to the entry plus 25 points, reducing the position size proportionally. This cap ensures that no single trade can risk more than MAX_RISK_PER_TRADE regardless of the entry and sweep combination.

26.3 Take Profit Levels

Two take profit levels are used. TP1 is set at 1:1 reward relative to the stop distance, meaning the distance from entry to TP1 equals the distance from entry to stop. TP2 is set at 2:1 reward. With a 25-point stop and order block entry, TP1 is 25 points in profit and TP2 is 50 points in profit per contract, representing \$50 and \$100 per contract on MNQ.

The two-target approach serves a specific function for prop firm compliance. TP1 at 1:1 converts the position to break-even as soon as it is hit, removing all downside risk. The subsequent move to TP2 is fully funded by the market. This dramatically reduces the occurrence of losses after an initial successful move, which is the most psychologically damaging trade outcome (giving back a winner). In the backtest, every trade that hit TP1 also hit TP2, because the sweep reversal model tends to produce sustained directional moves rather than shallow bounces.

26.4 Break-Even Stop Management

Immediately after TP1 is hit, the stop loss is moved to the entry price. This is the break-even management rule. If the market reverses after TP1, the position is closed at entry for zero P&L; (excluding spread). The original risk has been eliminated. The bot records this outcome as WIN_TP1_BE to distinguish it from a full win (TP2 hit) and from a loss. In the backtest, no WIN_TP1_BE outcomes occurred, meaning every trade that hit TP1 continued to TP2. This is expected given the institutional momentum behind valid sweep reversals, but will not always be the case in live trading.

26.5 Automated Execution Status

The full automated execution pipeline requires Tradovate API integration, implemented in the bot module's `api_client.py`. The API client supports placing bracket orders (entry + stop + TP1 + TP2 as a single order group), monitoring fill status, and modifying the stop to break-even when TP1 is detected as filled. This functionality is fully coded but requires an active funded account with API trading enabled to activate. During the evaluation phase, the bot operates in signal-only mode, displaying alerts for manual execution.

27. Future Research Directions and System Roadmap

The current system is a functional, validated first version. This section describes the research directions and planned improvements that would meaningfully extend its capabilities, along with the expected impact of each improvement on win rate, selectivity, and robustness.

27.1 Extended Backtest Validation

The most important near-term research priority is extending the backtest to cover 12 to 24 months of historical data across multiple market regimes: uptrending, downtrending, and ranging markets. The current 60-day window is dominated by a single bearish regime, which prevents any statistical conclusion about long-side setups or mean-reverting market behavior. A multi-year backtest would provide confidence intervals around the win rate estimates that are narrow enough to be actionable.

The expected finding is that win rate will be somewhat lower across a larger, more diverse sample. This is not a failure of the system but a correction for small-sample optimism. A win rate of 65 to 75 percent across a 24-month multi-regime backtest would still represent a highly profitable system with a strong theoretical foundation.

27.2 Long-Side Signal Validation

All nine trades in the current backtest were short. This is directionally correct for a bearish market, but it means the long-side logic has zero empirical validation. Long-side sweep setups occur when NQ sweeps below the Asia Low and reverses. These setups are theoretically identical to short setups in their structural requirements but may have different empirical characteristics: different average sweep depths, different MSS displacement patterns, or different relationships with London direction.

Validating long-side signals specifically requires a dataset that includes sustained uptrending periods. The 2023 and early 2024 NQ bull market would provide rich long-side signal history. Running the current system against that data with long-side signals enabled and tracking performance separately from short-side performance would identify whether any long-specific filters are needed.

27.3 Bayesian Threshold Adaptation

The current bot memory module updates thresholds based on point estimates of win rate with a hard minimum sample size of eight trades. A more sophisticated approach would use a Bayesian beta-binomial model to represent the uncertainty around the win rate estimate. With this model, the threshold adjustment would reflect not just the observed win rate but also the confidence in that estimate. Early trades would cause only small adjustments (high uncertainty), while after 20 or 30 trades in a category, the threshold would converge toward the true underlying rate.

This improvement would allow the system to begin adapting its thresholds from the very first live trade rather than waiting for the eight-trade minimum sample, while still avoiding over-adjustment during the noisy early period. The Python implementation would require only the `scipy.stats.beta` distribution, already a dependency, and minimal changes to `bot_memory.py`.

27.4 Multi-Instrument Application

The Asia Session Sweep model is theoretically applicable to any instrument that has a meaningful overnight session with a definable range and institutional participation at the US open. Candidates for expansion include: ES (E-mini S&P; 500, currently used only as SMT reference), RTY (Russell 2000 Micro, high beta), crude oil (CL, strong Asia range dynamics), and EURUSD (the London session plays the role of the New York open for forex). Each instrument would require separate empirical validation of sweep depth thresholds and DOW profiles.

27.5 Opening Range Refinement

The opening range opposed condition (point 7 of 9) is currently computed from the 9:30 to 10:00 AM candle sequence. ICT literature suggests that the 9:30 to 9:50 AM window is the more reliable manipulation window, and that the 9:50 to 10:00 AM window often begins to show the real direction. Refining the opening range window to 9:30 to 9:50 AM and adding a separate condition for the 9:50 AM directional confirmation would add one potential scoring point and improve the quality of the opening range analysis.

27.6 Intraday VWAP Bands

The current VWAP check is binary: price is either above or below VWAP. VWAP also has standard deviation bands (VWAP +/- 1 SD, +/- 2 SD) that define institutional value zones. A signal that fires with price at VWAP -2 SD (in the deep value zone for a long setup) has a stronger mean-reversion thesis than a signal at VWAP -0.5 SD. Adding a VWAP band condition that awards a point when price is in the deep value zone at signal time would refine the existing VWAP scoring from binary to three-level.

Improvement	Priority	Expected Impact	Complexity
24-month backtest validation	High	Statistical confidence	Low
Long-side signal validation	High	Complete DOW/direction matrix	Low
Bayesian threshold adaptation	Medium	Faster live adaptation	Medium
Multi-instrument application	Low	Diversified signal stream	High
Opening range refinement	Medium	+1 scoring point quality	Low
VWAP band conditions	Medium	VWAP scoring refinement	Low

28. Conclusion

The Asia Session Sweep System represents a practical implementation of institutional liquidity theory applied to equity index futures trading under the constraints of a proprietary trading firm evaluation. The system's defining characteristics are its extreme selectivity and its multi-layer validation architecture: a signal must pass through sweep detection, SMT cross-instrument confirmation, MSS structural validation, nine-point confluence scoring, smart adaptive filtering, and news calendar review before a trade is authorized.

The backtest iteration process demonstrated that targeted, data-driven filter improvements can dramatically improve win rate without destroying profitability: the three-round iteration moved the system from 40 percent to 89 percent win rate while maintaining a positive P&L trajectory. The key empirical findings were: sweep depth below 20 points has zero predictive value, Tuesday and Thursday require near-perfect confluence to avoid institutional manipulation traps, and SMT divergence is a reliable signal of Judas Swing conditions that should be hard-blocked.

The most significant limitation is the small backtest sample (nine trades) and the directional bias of the test period. Extended validation over multiple market regimes and directions is necessary before strong statistical conclusions can be drawn. The system's theory is sound and the empirical results are consistent with that theory, but small-sample statistical uncertainty remains.

The system is designed for ongoing improvement through its persistent bot memory module, which continuously updates threshold calibrations from live trade outcomes. As the system accumulates real trading history across diverse market conditions, the adaptive thresholds will replace the backtest-derived baselines with live-calibrated values, progressively improving its alignment with current market structure.

The ultimate objective is a fully automated system that executes bracket orders through the Tradovate API, moves stops to break-even at TP1, and closes positions at TP2 without manual intervention. The detection, scoring, and risk management infrastructure for this is complete. Activation requires only a funded account with active API access.

Appendix A: Glossary

Term	Definition
Asia Range	The high and low formed during the 8 PM to midnight EST session in equity index futures.
Drawdown	The decline from a peak account balance to a subsequent trough.
FVG	Fair Value Gap. A price imbalance where a candle's range does not overlap with the candle two bars prior.
ICT	Inner Circle Trader. A trading methodology developed by Michael J. Huddleston.
Judas Swing	ICT term for Tuesday's fake directional move designed to collect stops before the real weekly direction begins Wednesday.
Liquidity	Resting limit and stop orders in the market. Institutions target areas of high liquidity to fill large orders.
Liquidity Sweep	A price move that extends beyond a key level, collects stop orders there, then reverses sharply.
MNQ	Micro E-mini Nasdaq-100 futures. \$2.00 per index point per contract.
MSS	Market Structure Shift. A price close beyond a prior swing high or low after a sweep, confirming reversal.
NQ	E-mini Nasdaq-100 futures. \$20.00 per index point per contract. MNQ is one-tenth the size.
OB	Order Block. The last opposing candle before an institutional displacement move. Used as limit entry.
OTE	Optimal Trade Entry. The 61.8 to 78.6 percent Fibonacci retracement zone after a sweep.
Power of Three	ICT weekly model: Accumulation (Mon), Manipulation (Tue), Distribution (Wed-Fri).
SMT	Smart Money Technique. Cross-instrument divergence check comparing NQ and ES sweeps.
VWAP	Volume Weighted Average Price. A benchmark price that weights each tick by its volume.
VIX	CBOE Volatility Index. Measures expected 30-day implied volatility in the S&P 500.

Appendix B: 2026 Economic Calendar (Hardcoded Events)

The following events are hardcoded in strategy/market_context.py. All times are 8:30 AM EST unless otherwise noted. Blackout windows: 20 minutes before and 15 minutes after each event. Session-wide score penalty: +2 for all listed events, +1 for all Thursdays (jobless claims, not listed individually below).

Date	Event	Impact
Jan 2	NFP	Extreme
Jan 15	CPI	Extreme
Jan 29	FOMC (2:00 PM)	Extreme
Feb 6	NFP	Extreme
Feb 12	CPI	Extreme
Mar 6	NFP	Extreme
Mar 12	CPI	Extreme
Mar 19	FOMC (2:00 PM)	Extreme
Apr 3	NFP	Extreme
Apr 10	CPI	Extreme
May 1	NFP	Extreme
May 7	FOMC (2:00 PM)	Extreme
May 13	CPI	Extreme
Jun 5	NFP	Extreme
Jun 11	CPI	Extreme
Jun 18	FOMC (2:00 PM)	Extreme
Jul 2	NFP	Extreme
Jul 9	CPI	Extreme
Jul 30	FOMC (2:00 PM)	Extreme
Aug 7	NFP	Extreme
Aug 12	CPI	Extreme
Sep 4	NFP	Extreme
Sep 9	CPI	Extreme
Sep 17	FOMC (2:00 PM)	Extreme
Oct 2	NFP	Extreme

Oct 13	CPI	Extreme
Nov 5	FOMC (2:00 PM)	Extreme
Nov 6	NFP	Extreme
Nov 12	CPI	Extreme
Dec 4	NFP	Extreme
Dec 10	CPI + FOMC (2:00 PM)	Extreme

In addition to the above scheduled events, every Thursday carries a built-in +1 score penalty and a hard blackout from 8:10 AM to 8:45 AM EST for Initial Jobless Claims. This rule applies every single Thursday regardless of whether any other event is scheduled. The economic calendar should be updated annually with the following year's confirmed event dates before the start of each new calendar year.